Relations

Make structure explicit. Accept it independently.

ELI16 · THE IDEA IN PLAIN LANGUAGE

Labelled links between records: who depends on what, which role each endpoint plays, and which links were accepted.

Why it matters: A byte store cannot answer structural questions. Explicit ordered roles allow traversal without hiding semantics in payload conventions.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

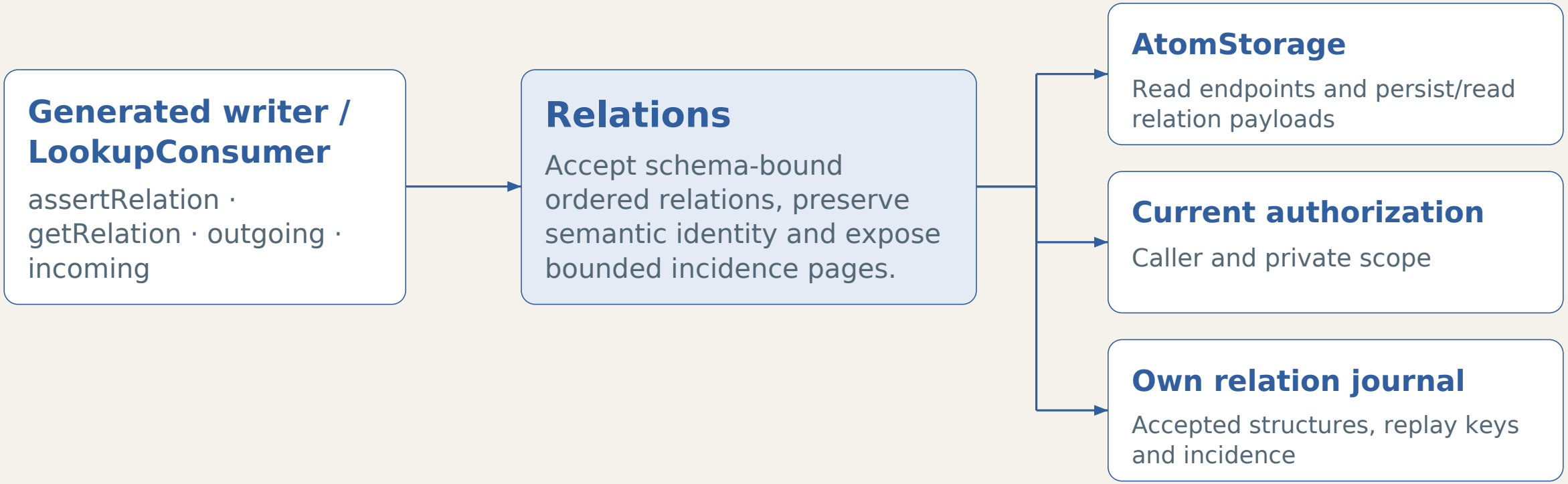
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	**Relations**	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Accept schema-bound ordered relations, preserve semantic identity and expose bounded incidence pages.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Accepted structural ledger

Schema, direction, ordered roles and acceptance sequence

Replay + incidence

Semantic duplicate handling and fixed-head traversal

OUTSIDE THIS RESPONSIBILITY

Unaccepted payloads

An orphan atom is not an accepted or traversable relation

Truth and general graph search

No inference, ontology engine, transitive query or hidden graph backend

schema-ordered-relations/2 · contract 2.0.0 · fresh stores

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
assertRelation	Validate structure, persist payload, accept	Semantic identity excludes transport key
getRelation	Read an accepted structure	Reject orphan or malformed relation payload
outgoing	Page origin-role incidence	Fixed head and bound cursor
incoming	Page destination-role incidence	Reverse incidence; symmetric schemas explicit

128 accepted relations; 512 replay keys; pages ≤8; finite schema set.

A worked example through the boundary

ACTUAL NATIVE HIVE LOOKUP · generated private corpus

01 · STEP

Discover root

Search returns a task reference.

02 · STEP

Follow outgoing

Read its requires relations at a fixed head.

03 · STEP

Inspect roles

Resolve dependent → prerequisite identities.

04 · STEP

Verify incoming

Confirm reverse incidence; retrieve payloads.

What the example establishes

A relation becomes visible when its owner accepts it.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Structural identity

Type/schema/direction/ordered roles/scope define meaning.

I2

One incidence

Self-links and repeated endpoints do not duplicate the same incidence.

I3

Snapshot pagination

Cursor binds principal, scope, endpoint, direction and accepted head.

I4

Independent acceptance

Payload creation and relation acceptance remain separate commits.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Crash after payload, before acceptance	Orphan stays invisible; retry may accept it.	One real process-kill case
Crash after acceptance, before reply	Recover the accepted relation once.	One real process-kill case
Changed cursor / revoked authority	Fail explicitly; no cross-scope continuation.	Malformed/cursor/auth tests

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

4 / 4

Logical operations exercised

Contract + real-provider + native consumer

15

Requires edges in native demo

Five roots connected to three prerequisites

128

Accepted relations per fresh store

Declared cap; no throughput claim

The 15-edge example demonstrates one-hop traversal and reverse incidence. No isolated relation throughput, latency or large-graph scaling benchmark has been run.

PROPOSED BENCHMARK · NOT RUN

Proposed: vary fan-out, schema arity, repeated endpoints and page size; measure append and traversal p50/p95 with fixed-head correctness under concurrent appends.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	SQLite RelationOwner over real local DeTTa; actual authenticated Hive traversal.
FIXTURE / LOCAL TEST POLICY	Generated schemas/corpus, malformed responses and injected faults; independent storage alternative.
BLOCKED / UNSUPPORTED	No required M5 gate blocked; historical tuple Relations declaration is not silently upgraded.
UNRUN	Large graphs, distributed writers, multi-hop query planning, migration and graph-database substitution.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Extend schema vocabulary	Versioned role semantics and negative tests	No ambiguous direction or structural identity
Measure fan-out costs	Controlled degree/arity/page matrix	Stable pagination with bounded memory
Explore a second provider	Independent accepted-state implementation	Same contract outcomes through unchanged consumer

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

A relation becomes visible when its owner accepts it.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m5.py --environment <env> --chain-build <chain> --output <fresh>
```

The all-gates run separates fixtures, real storage, native Hive and legacy regressions. Keep generated private corpus and journals private.

`fusion/relations.py`
`fusion/lookup_contracts.py`
`fusion/lookup.py`
`tests/test_m5.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.